

FMUSER FUQ-4443 IP QAM Modulator 1024 IP Inputs 32 QAM Outputs DVB-C RS(204-188) 3 GE Ports Multiplexing Scrambling CATV Broadcast Headend Product Overview

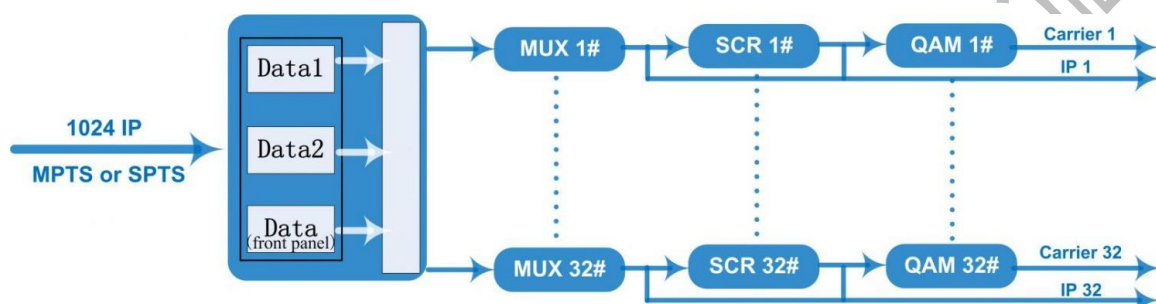


The FMUSER FUQ-4443 is an all-in-one IP QAM modulator integrating multiplexing, scrambling, and modulation, with three GE ports for up to 1024 IP inputs and a single RF interface for 32 QAM outputs. It processes signals by encoding and modulating them into DVB-C compliant formats. The device supports web-based network management and RS (204,188) encoding. Ideal for modern CATV broadcasting systems, it ensures scalability and high performance in cable TV headends and digital video distribution.

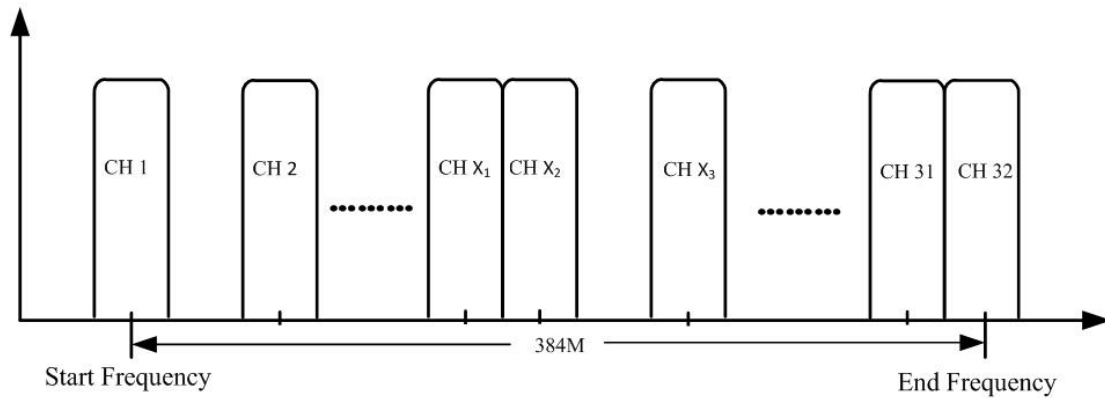
Key feature

- Supports up to 1024 IP inputs via three GE ports, each at 840Mbps.
- Outputs 32 non-adjacent QAM carriers compliant with DVB-C (EN 300 429) and ITU-T J.83 A/B.
- Enables PID remapping, PCR adjusting, CA filtering, and PSI/SI editing.
- Provides 32 IP outputs over UDP/RTP/RTSP via Data1 & Data2 ports.
- Includes web-based network management and RS (204,188) encoding for error correction.

Inner Principle Chart



Carrier Setting Illustration



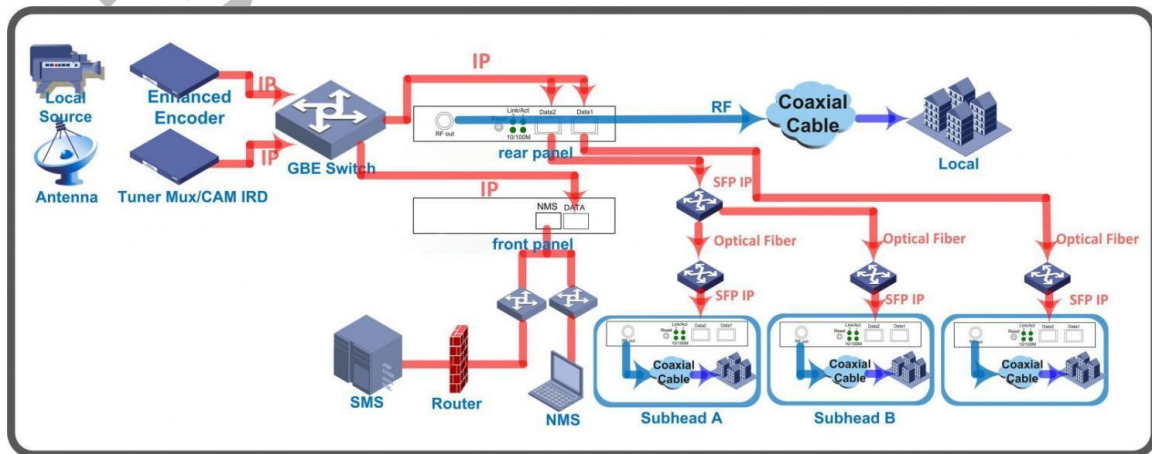
Specifications

Input	Input	Max 1024 IP input through 3 (front-panel Data port, Data 1 and Data 2) 100/1000M Ethernet Port (SFP interface optional). Each Data1 or Data 2 port can input maximum 512 IP, while front-panel Data port can input maximum 128 IP	Max 1024 IP input through 3 (front-panel Data port, Data 1 and Data 2) 100/1000M Ethernet Port (SFP interface optional). Each Data1 or Data 2 port can input maximum 512 IP, while front-panel Data port can input maximum 128 IP
Input	Transport Protocol	TS over UDP/RTP, unicast and multicast, IGMP V2/V3	TS over UDP/RTP, unicast and multicast, IGMP V2/V3
Input	Transmission Rate	Max 840Mbps for each GE input	Max 840Mbps for each GE input
Mux	Input Channel	1024	1024
Mux	Output Channel	32	32
Mux	Max PIDs	256 per channel	256 per channel
Mux	Functions	PID remapping(auto/manually optional)	PID remapping(auto/manually optional)
Mux	Functions	PCR accurate adjusting	PCR accurate adjusting
Mux	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating
Scrambling Parameters	Max simulscrypt CA	4	4

Scrambling Parameters	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters	Connection	Local/remote connection	Local/remote connection
Modulation Parameters	DVB-C Modulator Section	J.83A	Constellation : 16/32/64/128/256QAM
Modulation Parameters	DVB-C Modulator Section	J.83A	Bandwidth :8M
Modulation Parameters	DVB-C Modulator Section	J.83B	Constellation : 64QAM/ 256QAM
Modulation Parameters	DVB-C Modulator Section	J.83B	Bandwidth :6M
Modulation Parameters	QAM Channel	32 non-adjacent carrier	32 non-adjacent carrier
Modulation Parameters	Modulation Standard	EN300 429/ITU-T J.83A/B(DVB-C)	EN300 429/ITU-T J.83A/B(DVB-C)
Modulation Parameters	Symbol Rate	5.0~7.0Msps, 1ksps stepping	5.0~7.0Msps, 1ksps stepping
Modulation Parameters	Constellation	16, 32, 64 , 128, 256QAM	16, 32, 64 , 128, 256QAM
Modulation Parameters	FEC	RS (204, 188)	RS (204, 188)
RF Output	Interface	1 F typed output port for 32 carriers, 75Ω	1 F typed output port for 32 carriers, 75Ω
RF Output	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output	Output Level	-20dBm~+10dBm(87~117dbμV), 0.1dB stepping	-20dBm~+10dBm(87~117dbμV), 0.1dB stepping
RF Output	MER	≥ 40dB	≥ 40dB
TS output	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2 (Data1& Data2) 100/1000M Ethernet Ports	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2 (Data1& Data2) 100/1000M Ethernet Ports	32 IP output over UDP/RTP/RTSP, unicast/multicast, 2 (Data1& Data2) 100/1000M Ethernet Ports
System	Network management software (NMS) supporting	Network management software (NMS) supporting	Network management software (NMS) supporting

General	Demission	420mm×440mm×44.5 mm (WxLxH)	420mm×440mm×44.5 mm (WxLxH)
General	Temperature	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)
General	Power Supply	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz

System Connection Sample (IP Based)



About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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